NX220 SOC Analyst Project: CHECKER

Automatic SOC Attack Simulation and Logging System

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Student Code: s16

Unit: PERES25B

Program Code: NX220

Lecturer: Tzah Azoalis

1. Project Objective

The CHECKER project implements a Bash-based SOC manager tool. The tool discovers live IP addresses, displays controlled attack simulations, allows the operator to select an attack and target, executes the selected activity, and records the activity into a log file under /var/log.

The project uses safe lab-oriented simulations only. The selected activities demonstrate scanning, service identification, SMB exposure checking, random execution, invalid input handling, and logging in an authorized lab environment.

2. Lab Environment

Component	Value	Purpose
Attacker machine	Kali Linux - 192.168.47.136	Runs checker.sh and stores logs/results
Target machine	Windows - 192.168.47.129	Receives controlled scans in the lab
Network	VMware NAT - 192.168.47.0/24	Same subnet for attacker and target
Log file	/var/log/nx220_checker.log	Stores attack type, timestamp, target IP, and result file path
Results directory	~/nx220-checker-project/nx220-checker-results	Stores output files for each execution

3. Requirement Mapping

Requirement	Implementation	Evidence
At least 3 different attacks as separate functions	TCP Top Ports Scan, Service Version Detection, SMB Security Probe	Sections 6 to 8
Clear description for each attack	The menu prints a description for every attack before selection	Screenshot 2
Display available IP addresses	Nmap ping scan discovers live hosts on the local subnet	Screenshot 2
Allow specific or random attack	Manual choices 1 to 3 and random option R are supported	Section 9
Invalid input handling	Invalid attack selection terminates the program	Screenshot 2
Specify target IP or random target	Manual IP input and random target option R are supported	Section 9
Log selected activity in /var/log	Each execution is recorded in /var/log/nx220_checker.log	Sections 6 to 10

4. Script Preparation Evidence

Screenshot 1 - Script permissions

```
┌──(david㉿Kali)-[~/nx220-checker-project]
└─$ cd ~/nx220-checker-project
chmod +x checker.sh
ls -l checker.sh
-rwxrwxr-x 1 david david 7805 Jun 12 15:42 checker.sh
```

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checker.sh has executable permissions.

5. Network and Menu Evidence

Screenshot 2 - IP discovery, attack menu, and invalid input handling

```
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[*] Checking required tools ...
[OK] nmap found
[OK] curl found
[OK] ip found
[OK] awk found
[OK] sort found
[OK] shuf found
[OK] tee found

[*] Default interface: eth0
[*] Detected network: 192.168.47.136/24
[*] Discovering live IP addresses with nmap ping scan ...

192.168.47.1
192.168.47.2
192.168.47.129
192.168.47.136
192.168.47.254

[*] Total live IPs detected: 5
[*] Discovery file: /home/david/nx220-checker-project/nx220-checker-results/discovered_ips.txt

Available attack simulations:

1) TCP Top Ports Scan
   Description: Scans the target for common open TCP ports using Nmap.

2) Service Version Detection
   Description: Detects service versions on common ports using Nmap.

3) SMB Security Probe
   Description: Checks SMB exposure and security settings on TCP port 445 using Nmap scripts.

R) Random attack

Select attack number [1-3] or R for random: 9
[ERROR] Invalid attack selection. Terminating program.
```

The script discovers live IP addresses, displays the updated attack menu, and terminates on invalid input.

Screenshot 3 - Kali to Windows connectivity

```
┌──(david㉿Kali)-[~/nx220-checker-project]
└─$ ip -br addr
ip route
ip route get 192.168.47.129
ping -c 3 192.168.47.129
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0             UP             192.168.47.136/24 fe80::20c:29ff:fe39:a24d/64
default via 192.168.47.2 dev eth0 proto dhcp src 192.168.47.136 metric 100
192.168.47.0/24 dev eth0 proto kernel scope link src 192.168.47.136 metric 100
192.168.47.129 dev eth0 src 192.168.47.136 uid 1000
    cache
PING 192.168.47.129 (192.168.47.129) 56(84) bytes of data.
64 bytes from 192.168.47.129: icmp_seq=1 ttl=128 time=0.435 ms
64 bytes from 192.168.47.129: icmp_seq=2 ttl=128 time=0.474 ms
64 bytes from 192.168.47.129: icmp_seq=3 ttl=128 time=0.498 ms

--- 192.168.47.129 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2054ms
rtt min/avg/max/mdev = 0.435/0.469/0.498/0.025 ms
```

Kali can route to and ping the Windows target 192.168.47.129.

6. Attack 1 - TCP Top Ports Scan

The script executed a TCP top ports scan against the Windows target 192.168.47.129. The scan identified Windows-related services on ports 135, 139, and 445.

Screenshot 4 - Attack 1 result

```
┌──(david㉿Kali)-[~/nx220-checker-project]
└─$ cd ~/nx220-checker-project

echo "===== Latest TCP scan result file ====="
ls -t nx220-checker-results/tcp_top_ports_*.txt | head -n 1

echo
echo "===== File content ====="
cat "$(ls -t nx220-checker-results/tcp_top_ports_*.txt | head -n 1)"
===== Latest TCP scan result file =====
nx220-checker-results/tcp_top_ports_192.168.47.129_20260612_202619.txt

===== File content =====
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 20:26 +0300
Nmap scan report for 192.168.47.129
Host is up (0.00067s latency).
Not shown: 27 closed tcp ports (conn-refused)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 0.54 seconds
```

Nmap TCP scan result against the Windows target.

Screenshot 5 - Attack 1 log entry

```
┌──(david㉿Kali)-[~/nx220-checker-project]
└─$ sudo grep "attack=TCP_TOP_PORTS_SCAN" /var/log/nx220_checker.log | grep "target=192.168.47.129" | tail -n 1 | s
ed 's/ | /\n/g'
2026-06-14 12:52:13
student=David Rozi
unit=PERES25B
program=NX220
attack=TCP_TOP_PORTS_SCAN
target=192.168.47.129
output=/home/david/nx220-checker-project/nx220-checker-results/tcp_top_ports_192.168.47.129_20260614_125213.txt
```

Clean log entry for TCP_TOP_PORTS_SCAN.

7. Attack 2 - Service Version Detection

The script executed an Nmap service version detection scan against the Windows target 192.168.47.129. The scan identified Microsoft RPC, NetBIOS, and SMB-related services.

Screenshot 6 - Attack 2 result

```
┌──(david㉿Kali)-[~/nx220-checker-project]
└─$ cd ~/nx220-checker-project

echo "===== Latest service version result file ====="
ls -t nx220-checker-results/service_version_*.txt | head -n 1

echo
echo "===== File content ====="
cat "$(ls -t nx220-checker-results/service_version_*.txt | head -n 1)"
===== Latest service version result file =====
nx220-checker-results/service_version_192.168.47.129_20260613_234850.txt

===== File content =====
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-13 23:48 +0300
Nmap scan report for 192.168.47.129
Host is up (0.0013s latency).

PORT     STATE  SERVICE       VERSION
21/tcp   closed ftp
22/tcp   closed ssh
23/tcp   closed telnet
25/tcp   closed smtp
53/tcp   closed domain
80/tcp   closed http
110/tcp  closed pop3
111/tcp  closed rpcbind
135/tcp  open   msrpc         Microsoft Windows RPC
139/tcp  open   netbios-ssn   Microsoft Windows netbios-ssn
143/tcp  closed imap
443/tcp  closed https
445/tcp  open   microsoft-ds?
993/tcp  closed imaps
995/tcp  closed pop3s
1723/tcp closed pptp
3306/tcp closed mysql
3389/tcp closed ms-wbt-server
5900/tcp closed vnc
8080/tcp closed http-proxy
MAC Address: 00:0C:29:19:CB:BB (VMware)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.50 seconds
```

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Service version detection results for the Windows target.

Screenshot 7 - Attack 2 log entry

```
┌──(david㉿Kali)-[~/nx220-checker-project]
└─$ sudo grep "attack=SERVICE_VERSION_DETECTION" /var/log/nx220_checker.log | grep "target=192.168.47.129" | tail -
n 1 | sed 's/ | /\n/g'
2026-06-14 12:50:20
student=David Rozi
unit=PERES25B
program=NX220
attack=SERVICE_VERSION_DETECTION
target=192.168.47.129
output=/home/david/nx220-checker-project/nx220-checker-results/service_version_192.168.47.129_20260614_125020.txt
```

Clean log entry for SERVICE_VERSION_DETECTION.

8. Attack 3 - SMB Security Probe

The script executed an SMB security probe against the Windows target 192.168.47.129. The scan focused on TCP port 445 and identified supported SMB dialects.

Screenshot 8 - Attack 3 result

```
┌──(david㉿Kali)-[~/nx220-checker-project]
└─$ cd ~/nx220-checker-project

echo "===== Latest SMB security probe result file ====="
ls -t nx220-checker-results/smb_security_probe_*.txt | head -n 1

echo
echo "===== File content ====="
cat "$(ls -t nx220-checker-results/smb_security_probe_*.txt | head -n 1)"
===== Latest SMB security probe result file =====
nx220-checker-results/smb_security_probe_192.168.47.129_20260614_001218.txt

===== File content =====
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-14 00:12 +0300
Nmap scan report for 192.168.47.129
Host is up (0.00053s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds
MAC Address: 00:0C:29:19:CB:BB (VMware)

Host script results:
| smb-protocols:
|   dialects:
|     2.0.2
|     2.1
|     3.0
|     3.0.2
|_    3.1.1

Nmap done: 1 IP address (1 host up) scanned in 1.42 seconds
```

SMB security probe output showing SMB on TCP port 445 and supported dialects.

Screenshot 9 - Attack 3 log entry

```
┌──(david㉿Kali)-[~/nx220-checker-project]
└─$ sudo grep "attack=SMB_SECURITY_PROBE" /var/log/nx220_checker.log | grep "target=192.168.47.129" | tail -n 1 | s
ed 's/ | /\n/g'
2026-06-14 12:51:07
student=David Rozi
unit=PERES25B
program=NX220
attack=SMB_SECURITY_PROBE
target=192.168.47.129
output=/home/david/nx220-checker-project/nx220-checker-results/smb_security_probe_192.168.47.129_20260614_125107.tx
t
```

Clean log entry for SMB_SECURITY_PROBE.

9. Random Attack and Random Target

The script supports random attack selection and random target selection by entering R. In this execution, the script randomly selected attack 2 and target 192.168.47.129, then completed the scan and saved the output.

Screenshot 10 - Random selection and execution result

```
Select attack number [1-3] or R for random: R
[*] Random attack selected: 2

Detected IP addresses:
1) 192.168.47.1
2) 192.168.47.2
3) 192.168.47.129
4) 192.168.47.136
5) 192.168.47.254

Enter target IP manually or type R for random target: R
[*] Random target selected: 192.168.47.129

[*] Attack Simulation: Service Version Detection
[*] Target: 192.168.47.129
[*] Output: /home/david/nx220-checker-project/nx220-checker-results/service_version_192.168.47.129_20260614_133738.
txt

Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-14 13:37 +0300
Nmap scan report for 192.168.47.129
Host is up (0.00040s latency).

PORT     STATE  SERVICE       VERSION
21/tcp   closed ftp
22/tcp   closed ssh
23/tcp   closed telnet
25/tcp   closed smtp
53/tcp   closed domain
80/tcp   closed http
110/tcp  closed pop3
111/tcp  closed rpcbind
135/tcp  open   msrpc         Microsoft Windows RPC
139/tcp  open   netbios-ssn   Microsoft Windows netbios-ssn
143/tcp  closed imap
443/tcp  closed https
445/tcp  open   microsoft-ds?
993/tcp  closed imaps
995/tcp  closed pop3s
1723/tcp closed pptp
3306/tcp closed mysql
3389/tcp closed ms-wbt-server
5900/tcp closed vnc
8080/tcp closed http-proxy
MAC Address: 00:0C:29:19:CB:BB (VMware)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

The operator selected R for both attack and target; the script randomly selected Service Version Detection and target 192.168.47.129.

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Screenshot 11 - Random execution completed

```
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.11 seconds

════════════════════════════════════════════════════════════
Execution completed.
Log file:
/var/log/nx220_checker.log

Results directory:
/home/david/nx220-checker-project/nx220-checker-results

Last log entries:
2026-06-14 12:48:53 | student=David Rozi | unit=PERES25B | program=NX220 | attack=TCP_TOP_PORTS_SCAN | target=192.1
68.47.129 | output=/home/david/nx220-checker-project/nx220-checker-results/tcp_top_ports_192.168.47.129_20260614_12
4853.txt
2026-06-14 12:50:20 | student=David Rozi | unit=PERES25B | program=NX220 | attack=SERVICE_VERSION_DETECTION | targe
t=192.168.47.129 | output=/home/david/nx220-checker-project/nx220-checker-results/service_version_192.168.47.129_20
260614_125020.txt
2026-06-14 12:51:07 | student=David Rozi | unit=PERES25B | program=NX220 | attack=SMB_SECURITY_PROBE | target=192.1
68.47.129 | output=/home/david/nx220-checker-project/nx220-checker-results/smb_security_probe_192.168.47.129_202606
14_125107.txt
2026-06-14 12:52:13 | student=David Rozi | unit=PERES25B | program=NX220 | attack=TCP_TOP_PORTS_SCAN | target=192.1
68.47.129 | output=/home/david/nx220-checker-project/nx220-checker-results/tcp_top_ports_192.168.47.129_20260614_12
5213.txt
2026-06-14 13:37:38 | student=David Rozi | unit=PERES25B | program=NX220 | attack=SERVICE_VERSION_DETECTION | targe
t=192.168.47.129 | output=/home/david/nx220-checker-project/nx220-checker-results/service_version_192.168.47.129_20
260614_133738.txt
════════════════════════════════════════════════════════════
```

The random execution completed and the activity was logged.

10. Final Verification

The final verification section confirms that the main attacks were executed against the Windows target and that result files were generated. It also confirms that the final submission script file was created with the correct filename.

Screenshot 12 - Final log summary

```
┌──(david㉿Kali)-[~/nx220-checker-project]
└─$ cd ~/nx220-checker-project

echo "═══ Final CHECKER log summary ═══"

sudo grep -E "attack=TCP_TOP_PORTS_SCAN|attack=SERVICE_VERSION_DETECTION|attack=SMB_SECURITY_PROBE" /var/log/nx220_
checker.log \
| grep "target=192.168.47.129" \
| tail -n 3 \
| sed 's/ | /\n/g'
═══ Final CHECKER log summary ═══
2026-06-14 12:51:07
student=David Rozi
unit=PERES25B
program=NX220
attack=SMB_SECURITY_PROBE
target=192.168.47.129
output=/home/david/nx220-checker-project/nx220-checker-results/smb_security_probe_192.168.47.129_20260614_125107.tx
t
2026-06-14 12:52:13
student=David Rozi
unit=PERES25B
program=NX220
attack=TCP_TOP_PORTS_SCAN
target=192.168.47.129
output=/home/david/nx220-checker-project/nx220-checker-results/tcp_top_ports_192.168.47.129_20260614_125213.txt
2026-06-14 13:37:38
student=David Rozi
unit=PERES25B
program=NX220
attack=SERVICE_VERSION_DETECTION
target=192.168.47.129
output=/home/david/nx220-checker-project/nx220-checker-results/service_version_192.168.47.129_20260614_133738.txt
```

Filtered log summary for final executions against the Windows target.

Screenshot 13 - Final result files

```
┌──(david㉿Kali)-[~/nx220-checker-project]
└─$ cd ~/nx220-checker-project

echo "═════ Final result files for Windows target ═════"

ls -1t nx220-checker-results/*192.168.47.129*.txt | head -n 6
═════ Final result files for Windows target ═════
nx220-checker-results/service_version_192.168.47.129_20260614_133738.txt
nx220-checker-results/tcp_top_ports_192.168.47.129_20260614_125213.txt
nx220-checker-results/smb_security_probe_192.168.47.129_20260614_125107.txt
nx220-checker-results/service_version_192.168.47.129_20260614_125020.txt
nx220-checker-results/tcp_top_ports_192.168.47.129_20260614_124853.txt
nx220-checker-results/smb_security_probe_192.168.47.129_20260614_001218.txt
```

Clean list of result files related to the Windows target.

Screenshot 14 - Final script file

```
┌──(david㉿Kali)-[~/nx220-checker-project]
└─$ cd ~/nx220-checker-project

cp checker.sh PERES25B.s16.nx220.sh
chmod +x PERES25B.s16.nx220.sh

bash -n PERES25B.s16.nx220.sh && echo "[OK] final submission script is valid"

ls -l PERES25B.s16.nx220.sh
[OK] final submission script is valid
-rwxrwxr-x 1 david david 7618 Jun 14 13:40 PERES25B.s16.nx220.sh
```

The final Bash script file PERES25B.s16.nx220.sh exists, is executable, and passes Bash syntax validation.

11. Submission Checklist

- Export this document to PDF as PERES25B.s16.nx220.pdf.
- Submit the Bash script file: PERES25B.s16.nx220.sh.
- Submit the PDF file: PERES25B.s16.nx220.pdf.
- Do not submit the DOCX file if the platform accepts only .sh and .pdf.
- For GitHub, include README.md, the Bash script, the final PDF, and sanitized screenshots if needed.